

Stationary source emissions — Determination of volume concentration of oxygen (O₂) — Reference method — Paramagnetism

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National foreword

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The UK participation in its preparation was entrusted by Technical Committee EH/2, Air quality, to Subcommittee EH/2/1, Stationary source emissions, which has the responsibility to:

- aid enquirers to understand the text;
- present to the responsible international/European committee any enquiries on the interpretation, or proposals for change, and keep UK interests informed;
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Emissions de sources fixes - Détermination de la concentration volumique en oxygène (O₂) - Méthode de référence: Paramagnétisme

Emissionen aus stationären Quellen - Bestimmung der Volumenkonzentration von Sauerstoff - Referenzverfahren - Paramagnetismus

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Foreword

This European Standard (EN 14789:2005) has been prepared by Technical Committee CEN/TC 264 "Air quality", the secretariat of which is held by DIN.

This European Standard shall be given the status of a national standard, either by publication of an identical text or by endorsement, at the latest by May 2006, and conflicting national standards shall be withdrawn at the latest by May 2006.

This European Standard has been prepared under a mandate given to CEN by the European Commission and the European Free Trade Association, and supports essential requirements of EU Directive(s).

For relationship with EU Directive(s), see informative Annex ZA, which is an integral part of this European Standard.

According to the CEN/CENELEC Internal Regulations, the national standards organizations of the following countries are bound to implement this European Standard: Austria, Belgium, Cyprus, Czech Republic, Denmark, Estonia, Finland, France, Germany, Greece, Hungary, Iceland, Ireland, Italy, Latvia, Lithuania, Luxembourg, Malta, Netherlands, Norway, Poland, Portugal, Slovakia, Slovenia, Spain, Sweden, Switzerland and United Kingdom.

1 Scope

This European Standard describes the paramagnetic method, including the sampling and the gas conditioning system, to determine the oxygen concentrations in flue gases emitted to the atmosphere from ducts and stacks.

This European Standard is the Standard Reference Method (SRM) for periodic monitoring and for the calibration or control of Automatic Measuring Systems (AMS) permanently installed on a stack, for regulatory purposes or other purposes. To be used as the SRM, the user shall demonstrate that the performance characteristics of the method are better than the performance criteria defined in this European Standard and that the overall uncertainty of the method is less than $\pm 6,0$ % of the measured concentration.

NOTE When paramagnetism is the measurement principle used for AMS, reference should be made to EN 14181 and other relevant standards provided by CEN/TC 264.

An alternative method to this SRM may be used provided that the user can demonstrate equivalence according to the Technical Specification CEN/TS 14793, to the satisfaction of his national accreditation body or law.

This Standard Reference Method has been evaluated during field tests on waste incineration, co-incineration and large combustion installations. It has been validated for sampling periods of 30 min in the range: 5 % to 26 %. Oxygen concentration values, expressed in % volume, are used in order to allow emission measurements of pollutants to be standardised to the reference O₂ concentration and dry gas conditions required by the following Council Directives:

- Council Directive 2001/80/EC on the limitation of emissions of certain pollutants into the air from large combustion plants;
- Council Directive 2000/76/EC on waste incineration plants.

2 Normative references

The following referenced documents are indispensable for the application of this document. For dated references, only the edition cited applies. For undated references, the latest edition of the referenced document (including any amendments) applies.

CEN/TS 14793, *Stationary source emission – Intralaboratory validation procedure for an alternative method compared to a reference method*.

ENV 13005, *Guide to the expression of uncertainty in measurement*.

EN ISO 14956, *Air quality - Evaluation of the suitability of a measurement procedure by comparison with a required measurement uncertainty (ISO 14956:2002)*.

3 Terms and definitions

For the purposes of this European Standard, the following terms and definitions apply.

NOTE In this European Standard, the concentration of O₂ is expressed in % volume.

3.1

adjustment (of a measuring system)

operation of bringing a measuring system into a state of performance suitable for its use

[VIM 4.30]

3.2

ambient temperature

temperature of the air around the measuring device

3.3

automatic measuring system (AMS)

measuring system permanently installed on site for continuous monitoring of emissions

NOTE 1 An AM is a method which is traceable to a reference method.

NOTE 2 Apart from the analyser, an AMS includes facilities for taking samples (e.g. probe, sample gas lines, flow meters, regulators, delivery pumps) and for sample conditioning (e.g. dust filter, moisture removal devices, converters, diluters). This definition also includes testing and adjusting devices, that are required for regular functional checks.

[EN 14181]

3.4

calibration

statistical relationship between values of the measurand indicated by the measuring system (AMS) and the corresponding values given by the standard reference method (SRM) used during the same period of time and giving a representative measurement on the same sampling plane

NOTE The result of calibration permits to establish the relationship between the values of the SRM and the AMS (calibration function).

3.5

drift

difference between two zero (zero drift) or span readings (span drift) at the beginning and at the end of a measuring period

3.6

emission limit value (ELV)

emission limit value according to EU Directives on the basis of 30 min, 1 hour or 1 day

3.7

influence quantity

quantity that is not the measurand but that affects the result of the measurement

[adapted VIM 2.7]

NOTE Examples:

- ambient temperature;
- atmospheric pressure;
- presence of interfering gases in the flue gas matrix;
- pressure of the gas sample.

3.8

interference

negative or positive effect upon the response of the measuring system, due to a component of the sample that is not the measurand

3.9**lack of fit**

systematic deviation within the range of application between the measurement result obtained by applying the calibration function to the observed response of the measuring system measuring test gases and the corresponding accepted value of such test gases

NOTE 1 Lack of fit may be a function of the measurement result.

NOTE 2 The expression "lack of fit" is often replaced in everyday language by "linearity" or "deviation from linearity".

3.10**measurand**

particular quantity subject to measurement

[VIM 2.6]

3.11**measuring system**

complete set of measuring instruments and other equipment assembled to carry out specified measurements

[VIM 4.5]

3.12**performance characteristic**

one of the quantities (described by values, tolerances, range) assigned to equipment in order to define its performance

3.13**repeatability in the laboratory**

closeness of the agreement between the results of successive measurements of the same measurand carried out under the same conditions of measurement

NOTE 1 Repeatability conditions include:

- same measurement procedure;
- same laboratory;
- same measuring instrument, used under the same conditions;
- same location;
- repetition over a short period of time.

NOTE 2 Repeatability may be expressed quantitatively in terms of the dispersion characteristics of the results.

In this European Standard the repeatability is expressed as a value with a level of confidence of 95 %.

[VIM 3.6]

3.14

repeatability in the field

closeness of the agreement between the results of simultaneous measurements of the same measurand carried out with two sets of equipment under the same conditions of measurement

NOTE 1 These conditions include:

- same measurement procedure;
- two sets of equipment, the performances of which are fulfilling the requirements of the reference method, used under the same conditions;
- same location;
- implemented by the same laboratory;
- typically calculated on short periods of time in order to avoid the effect of changes of influence parameters (e.g 30 min).

NOTE 2 Repeatability may be expressed quantitatively in terms of the dispersion characteristics of the results.

In this European Standard, the repeatability under field conditions is expressed as a value with a level of confidence of 95 %.

3.15

reproducibility in the field

closeness of the agreement between the results of simultaneous measurements of the same measurand carried out with several sets of equipment under the same conditions of measurement

NOTE 1 These conditions are called field reproducibility conditions and include:

- same measurement procedure;
- several sets of equipment, the performances of which fulfil the requirements of the reference method, used under the same conditions;
- same location;
- implemented by several laboratories.

NOTE 2 Reproducibility may be expressed quantitatively in terms of the dispersion characteristics of the results.

In this European Standard, the reproducibility under field conditions is expressed as a value with a level of confidence of 95 %.

3.16

residence time in the measuring system

time period for the sampled gas to be transported from the inlet of the probe to the inlet of the measurement cell

3.17

response time

time interval between the instant when a stimulus is subjected to a specified abrupt change and the instant when the response reaches and remains within specified limits around its final steady value

NOTE By convention time taken for the output signal to pass from 0 % to 90 % of the final change.

[VIM 5.17]

3.18**sampling location**

specific area, close to the sampling plane, where the measurement devices are set up

3.19**sampling plane**

plane normal to the centreline of the duct at the sampling position

[EN 13284-1]

3.20**sampling point**

specific position on a sampling line at which a sample is extracted

[EN 13284-1]

3.21**span gas**

test gas used to adjust and check a specific point on the response line of the measuring system

NOTE This concentration is often chosen around 80 % of the upper limit of the range.

3.22**Standard Reference Method (SRM)**

measurement method recognised by experts and taken as a reference by convention, which gives, or is presumed to give, the accepted reference value of the concentration of the measurand (3.10) to be measured

3.23**uncertainty**

parameter associated with the result of a measurement, that characterises the dispersion of the values that could reasonably be attributed to the measurand

3.23.1**standard uncertainty u**

uncertainty of the result of a measurement expressed as a standard deviation u

3.23.2**combined uncertainty u_c**

standard uncertainty u_c attached to the measurement result calculated by combination of several standard uncertainties according to GUM

3.23.3**expanded uncertainty U**

quantity defining a level of confidence about the result of a measurement that may be expected to encompass a specific fraction of the distribution of values that could reasonably be attributed to a measurand

$$U = k \times u$$

NOTE In this European Standard, the expanded uncertainty is calculated with a coverage factor of $k=2$, and with a level of confidence of 95 %.

3.23.4**overall uncertainty U_c**

expanded combined standard uncertainty attached to the measurement result calculated according to GUM

$$U_c = k \times u_c$$

3.24**uncertainty budget**

calculation table combining all the sources of uncertainty according to EN ISO 14956 or ENV 13005 in order to calculate the overall uncertainty of the method at a specified value

4 Principle**4.1 General**

This European Standard describes the Standard Reference Method (SRM), based on the paramagnetism principle for sampling and determining oxygen O₂ concentration in flue gases emitted to atmosphere from ducts and stacks. The specific components and the requirements for the sampling system and the paramagnetic analyser are described. A number of performance characteristics, together with associated minimum performance criteria are specified for the analyser. These performance characteristics and the overall uncertainty of the method shall meet the performance criteria given in this European Standard. Requirements and recommendations for quality assurance and quality control are given for measurements in the field (see Table 1 in 7.2).

4.2 Measuring principle

The paramagnetic method is based on the principle that oxygen molecules are strongly attracted to a magnetic field. This property, known as paramagnetism, can be used for the selective measurement of oxygen in flue gases where the other constituents are either slightly or non-paramagnetic. The magnetic susceptibility or degree of magnetisation produced in a gas sample by a magnetic field is inversely proportional to its absolute temperature. A gas sample containing oxygen, when exposed to the combined effect of a magnetic gradient in a confined space, shall be constrained to flow in the direction of the magnetic field. The magnitude of this flow, other factors being equal, is dependent on the oxygen concentration in the gas sample induced flow. A number of devices, described in clause 6 have been developed to measure the paramagnetically induced flow.

Paramagnetic analysers are combined with an extractive sampling system and a gas conditioning system. A representative sample of gas is taken from the stack with a sampling probe and conveyed to the analyser through a sampling line and suitable gas conditioning system. The values from the analyser are recorded and/or stored by means of electronic data processing.

5 Description of measuring equipment - Sampling and sample gas conditioning systems**5.1 General**

A representative volume of flue gas (see 8.2) is extracted from the emission source for a fixed period of time at a controlled flow rate. A filter removes the dust in the sampled volume before the sampled gas is conditioned and passed to the analyser.

Two different sampling and conditioning configurations are available in order to avoid uncontrolled water condensation in the measuring system. Both approaches require that the user shall check that the dew point temperature is lower or equal to 4 °C at the outlet of the analyser. The user may correct the results for the remaining water content in order to report results on a dry basis (refer to the table of Annex A in EN 14790:2003).

These configurations are:

- configuration 1: removal of water vapour by condensation using a cooling system;
- configuration 2: removal of water vapour through elimination within a permeation drier.

Schematic diagrams of typical measuring systems are shown in Annex A.

It is important that all parts of the sampling equipment upstream of the analyser are made of materials that do not react with or absorb O₂. Except for the cooling system of configuration 1, the temperature of its components coming into contact with the gas, shall be maintained at a sufficiently high temperature to avoid any condensation.

Conditions and layout of the sampling equipment contribute to the overall uncertainty of the measurement. In order to minimise this contribution to the overall measurement uncertainty, the performance criteria for the sampling equipment and sampling conditions are given in sections 5.2 and 6.

5.2 Sampling line components

5.2.1 Sampling line

In order to access the representative measurement point(s) of the sampling plane, probes of different lengths and inner diameters may be used. The design and configuration of the probe used shall ensure the residence time of the sample gas within the probe is minimised in order to reduce the response time of the measuring system.

The procedure of sub-clause 8.2 shall be used when the operator suspects that the flue gas is inhomogeneous.

The probe may be marked before sampling in order to demonstrate that the representative measurement point(s) in the measurement plane has (have) been reached.

The line shall be made of a suitable, corrosion resistant material (e.g. stainless steel, borosilicate glass, ceramic, titanium, PTFE is only suitable for flue gas temperatures lower than 200 °C).

A seal-able connection may be installed on the probe in order to introduce test gases for adjustment.

5.2.2 Filter

The particle filter shall be made of an inert material (e.g. ceramic or sinter metal filter with an appropriate pore size). It shall be heated above the water or acid dew point. The particle filter shall be changed or cleaned periodically depending on the dust loading at the sampling site.

NOTE Overloading of the particle filter may increase the pressure drop in the sampling line.

5.2.3 Sample cooler or permeation drier

The sample cooler or permeation drier is used before the gas enters the analyser in order to separate water vapour from the flue gas. A maximum dew-point temperature of 4 °C shall not be exceeded at the outlet of the system.

NOTE The measured oxygen concentration, given by these sampling configurations, can be considered to be dry. However, the user may correct the results for the remaining water (refer to the Table of Annex A in EN 14790:2003).

5.2.4 Sample pump

When a pump is not an integral part of the paramagnetic analyser, an external pump is necessary to draw the sampled air through the apparatus. It shall be capable of operating according to the specified flow requirements of the manufacturer of the analyser and pressure conditions required for the reaction chamber. The pump shall be resistant to corrosion and consistent with the requirements of the analyser to which it is connected.

NOTE The quantity of sample gas required can vary between 15 l/h and 500 l/h, depending upon the analyser and the expected response time.

5.2.5 Secondary filter

The secondary filter is used to separate fine dust, with a pore size of 1 µm to 2 µm. It may be made in glass-fibre, sintered ceramic, stainless steel or PTFE.

NOTE No additional secondary filter is necessary when they are part of the analyser itself.

5.2.6 Flow controller and flow meter

This apparatus sets the required flow. A corrosion resistant material shall be used. The sample flow rate into the instrument shall be maintained according to the analyser manufacturer's requirements. A controlled pressure drop across restrictors is usually employed to maintain flow rate control into the analyser.

NOTE No additional flow controller or flow meter is necessary when they are part of the analyser itself.

6 Analyser equipment

Several variants of application of the paramagnetism principle are available. Some of them are described below:

— Thermo-magnetic:

Two separate chambers (reference and measuring chambers) are equipped with thermo-sensitive resistors, which form an assembly with a Wheatstone bridge. The measuring chamber is located in a magnetic field while the reference is not. When the concentration of oxygen increases, the flow in the measurement chamber is greater than the flow in the reference chamber. This creates a differential cooling effect on the resistors. The equilibrium of the Wheatstone bridge is restored by an increase of an electric current in the resistors and this current is proportional to the oxygen concentration.

— Magneto-mechanic:

In one measuring chamber, two permanent magnets create a non-homogenous magnetic field in which a very light float is suspended. When the gas is injected into the chamber, the oxygen molecules are attracted by the magnetic field. This in turn creates a differential partial pressure, which then alters the position of the float. An electromagnetic field is then applied to compensate for this change, while the strength of the applied compensatory field is proportional to the concentration of oxygen.

— Magneto-pneumatic:

The sampled gas and a reference gas circulate within an electromagnet. Each gas is subject to a modification of pressure that is proportional to the oxygen concentration. This pressure difference then alters the position of the membrane of a capacitor. The variation in capacitance allows the conversion of the pressure signal to an electric signal, which in turn is proportional to the partial pressure of oxygen.

7 Determination of the characteristics of the SRM : analyser, sampling and conditioning line

7.1 General

When this European standard is used as the SRM, the user shall demonstrate that:

- performance characteristics of the method are better than the minimum performance criteria given in Table 1; and
- overall uncertainty calculated by combining values of standard uncertainties associated to the performance characteristics given in Table 1 is less than $\pm 6,0$ % rel. of the measured value expressed on dry basis.

The values of the selected performance characteristics shall be evaluated by means of a laboratory test and a field test. An experienced laboratory recognised by the competent authority shall perform the laboratory tests according to the procedures described in the relevant CEN/TC 264 standards or technical specifications. The user shall perform the field tests.

The performance characteristics and performance criteria given in this European Standard do not necessarily correspond to those of an AMS. For AMS performance characteristics and performance criteria the user shall refer to the relevant standards provided by CEN/TC 264.

7.2 Relevant performance characteristics of the SRM and performance criteria

The uncertainty of the measured values given by an analyser is not only influenced by the performance characteristics of the analyser itself but also by:

- sampling system including conditioning;
- site specific conditions;
- calibration gas used.

Table 1 gives an overview of the minimum performance characteristics and performance criteria, which shall be determined during laboratory and field tests according to the relevant CEN procedures, and indicates values included in the calculation of the overall uncertainty.

Table 1 — Minimum performance characteristics of the SRM

Performance characteristic	Lab. Test for the analyser	Field test	Performance criterion	Performance characteristics to be included in calculation of overall uncertainty
Response time	X	X	≤ 200 s	
Detection limit	X		$\leq \pm 0,20$ % relative of the range	
Lack of fit	X		$\leq \pm 0,30$ % volume	X
Zero drift	X		$\leq \pm 0,20$ % volume/24 h	X
Span drift	X		$\leq \pm 0,20$ % volume/24 h	X
Sensitivity to atmospheric pressure	X		$\leq \pm 3,0$ % relative of the range for 2 kPa	X
Sensitivity to sample volume flow or sample pressure	X		^a	X
Sensitivity to ambient temperature	X		$\leq 0,30$ % volume/10 K	X
Sensitivity to electric voltage	X		$\leq \pm 0,10$ % volume/10 V	X
Interferents ^b	X		Total $\leq \pm 0,20$ % volume	X
Standard deviation of repeatability in laboratory at zero	X		$\leq \pm 0,20$ % relative of the range	X ^c
Standard deviation of repeatability in laboratory at span level	X		$\leq 0,40$ % relative of the range	X ^c
Losses and leakage in the sampling line and conditioning system		X	$\leq \pm 2,0$ % relative of the measured value	X

^a The tested volume flow range or pressure is defined in the manufacturer's recommendations.

^b Interferents that shall be tested are at least those given in Table 2.

The value of max algebraic sums of contributions to uncertainty producing positive effects on the result, (sum of contributions to uncertainty producing negative effects) shall be compared with the performance criterion.

^c Only one of both values shall be included in the calculation: the first possibility is to choose the repeatability standard deviation got from laboratory tests corresponding to the closest concentration to the actual concentration in stack, or the higher (relative) standard deviation of repeatability independently of the concentration measured in stack.

7.3 Establishment of the uncertainty budget

An uncertainty budget shall be established to determine if the analyser and its associated sampling system fulfil the requirements for a maximum allowable overall uncertainty.

The overall uncertainty for this method used as a reference shall be lower than 6,0 % of the measured value expressed on dry basis.

The principle of calculation of the overall uncertainty is based on the law of propagation of uncertainty laid down in ENV 13005:

- determine the standard uncertainties attached to the performance characteristics to be included in the calculation of the uncertainty budget by means of laboratory and field tests, and according to ENV 13005;
- calculate the uncertainty budget by combining all the standard uncertainties according to ENV 13005, including the uncertainty of the calibration gas and taking variations range of influence quantities and interferences of the specific site conditions into account. If these conditions are unknown, default values defined in Table 2 shall be applied;
- values of standard uncertainty that are less than 5 % of the maximum standard uncertainty can be neglected;
- calculate the overall uncertainty at the measured value, reported as a dry gas value.

Table 2 — Default variations ranges of influence quantities and values of interferences to be applied for the uncertainty budget

Influence quantity	Variations range on site
Atmospheric pressure	± 2 kPa
Sample volume flow or pressure variation	in accordance with the manufacturer's recommendations
Ambient temperature	± 15 °C
Electric voltage at span level	230 V $\pm$ 20 V
NO	300 mg NO/m ³
NO ₂	30 mg NO ₂ /m ³
CO ₂	15 % volume

An example of an uncertainty budget is given in Annex B.

8 Field operation

8.1 Sampling location

The sampling location chosen for the measurement devices and samplings shall be of sufficient size and construction, that a representative emission measurement can be made that is suitable for the measurement task and technically perfect is possible. In addition, the sampling location shall be chosen with regard to safety of the personnel, accessibility and availability of electrical power.

8.2 Sampling point(s)

The gas concentrations measured shall be representative of the average condition of the waste gas. In most cases, a single sampling point situated in the middle of the duct shall be selected. For larger ducts, this point may be situated close to the sampling port provided that there is no disturbance of the flow or concentration due to the influence of the sampling port.

However, when non-homogeneity of the flue gas is suspected, it shall be demonstrated if the flue gas is homogeneous or not. The homogeneity can be demonstrated with a continuous measurement of O₂ or CO₂ using a fixed and a moving probe. Sampling in non-homogeneous flue gases is required to follow relevant standards or Technical Specifications proposed by CEN/TC 264.

8.3 Choice of the measuring system

The following flue gas characteristics shall be determined, before the field campaign, so that an appropriate analyser and sampling line configuration (including conditioning unit) can be adopted:

- flue gas moisture content;
- temperature of exhaust gases;
- dust loading;
- expected concentration of potentially interfering substances, including at least the components listed in Table 2.

The sampling line should be as short as possible to avoid long response times. If necessary a bypass pump together with a heated filter appropriate to the dust loading shall be used.

The laboratory shall verify that the analyser is functioning in accordance with the required performance criteria fixed in this European Standard, and that the sampling line and conditioning unit are in good operational conditions before conducting field measurements.

8.4 Setting of the SRM on site

8.4.1 General

The complete measuring system, including the conditioning unit, the sampling line and the analyser, shall be connected according to the manufacturer's instructions and the nozzle of the probe placed at the representative point(s) in the duct (see 8.2).

The conditioning unit, sampling probe, filter, connection tube and analyser shall be stabilised at the required temperature. At the same time, a constant pressure shall be achieved in the measuring chamber of the analyser.

After pre-heating, the flow passing through the sampling system and the analyser shall be adjusted to the chosen flow rate to be used during measurement. This flow shall be maintained at a constant level (+/- 10 %).

When both the instrument and sampling system have been set-up, the proper functioning of the instrument and sampling system shall be checked. The results of these checks shall fulfil the requirements and limitations as set out by the manufacturer of the instrument as well as the requirements (such as materials used and so on) given in this European Standard. Compliance with the requirements of this European Standard shall be documented.

Any data recording, data processing and telemetry system used in conjunction with the measuring system shall be checked for proper functioning. If any components are changed, then these checks shall be repeated. All checks shall be documented. The time resolution of the data used to calculate the mean values shall be below or equal to 1 min.

8.4.2 Preliminary zero and span check, and adjustments

8.4.2.1 Test gases

8.4.2.1.1 Zero gas

The zero gas shall be free of oxygen ($< 0,05\%$ vol).

8.4.2.1.2 Span gases

The span gas used to adjust the analyser shall have a certified concentration of O_2 . The expanded uncertainty on the analytical certificate of the span gas shall be less than $\pm 2\%$. Ambient air, the concentration of which is $20,9\%$ vol O_2 , can also be used for adjustment of the analyser (uncertainty: $\pm 0,5\%$ rel). This air shall be clean and dried by passing through an appropriate agent (e.g. silicagel).

8.4.2.2 Adjustment of the analyser

At the beginning of the measuring period, zero and span gases or air are supplied to the analyser directly, without passing through the sampling system. Adjustments are made until the correct zero and span gas values are reached by the data sampling system and are stable:

- adjust the zero value;
- adjust the span;
- finally, check again zero to see if there is no significant changes (zero deviation lower than 2 times the repeatability at zero). If there is any problem, repeat the procedure.

8.4.2.3 Check of the sampling system

Before starting the measurement, one of these two following procedures shall be applied to check that there is no significant leakage in the sampling line:

- zero and span gas are supplied to the analyser through the sampling system, as close as possible to the nozzle (in front of the filter if possible). Differences on the readings shall be lower than 2% rel;
- check the sampling line for leakage according to the following procedure or any other relevant procedure.

Assemble the complete sampling system. Close the nozzle and switch on the pump. After reaching minimum pressure, read or measure the flow rate with an appropriate measuring device. The leak flow rate shall not exceed 2% of the expected sample gas flow rate used during measurement.

8.4.3 Zero and span checks after measurement

At the end of the measuring period and at least once a day, zero and span checks shall be performed at the inlet of the sampling system by supplying test gases. The information shall be documented. In case of deviation between checks after measurement and preliminary adjustments, values of deviation shall be indicated in the report.

If the span or zero drifts are bigger than 2% of the span value, it is necessary to correct both for zero and span drifts (see in Annex C a procedure of correction of data for drift effect).

The drift of zero and span shall be lower than 5% of the span value; otherwise, the results shall be rejected.

9 Ongoing quality control

9.1 General

Quality control is critically important in order to ensure that the uncertainty of the measured values is kept within the stated limits during extended continuous monitoring periods in the field. This means that maintenance, as well as zero and span adjustment procedures, shall be followed, as they are essential for obtaining accurate and traceable quality data.

9.2 Frequency of checks

Table 4 shows the minimum required frequency of checks. The laboratory shall implement the relevant European Standards for determination of performance characteristics.

Table 3 — Frequency of checks as SRM

Checks	Frequency	Action criteria
Cleaning or changing of particulate filters ^a at the sampling inlet and at the monitor inlet	Every campaign if needed ^a	
Regular maintenance of several parts of the monitor	As required by the manufacturer	As required
Linearity check	At least every year and after repair of the instrument	As required and when linearity > $\pm 0,30$ % volume
^a The particle filter shall be changed periodically depending on the dust loading at the sampling site and the filter housing shall be cleaned. Overloading of the particle filter may increase the pressure drop in the sampling line.		

10 Expression of results

The readings from the analyser are converted to concentrations using the appropriate calibration function and the results are expressed as percentages by volume.

Reference concentrations of O₂ defined in the European directives are:

- 11 % for incineration of waste in waste incinerators;
- 10 % for co-incineration of waste in cement kilns;
- 3 % for combustion of gas or liquid fuels;
- 6 % for combustion of solid fuels;
- 15 % for gas turbines.

11 Evaluation of the method in the field

The method has been evaluated during six field tests, on waste incineration, co-incineration and large combustion installations. Each test was performed by at least four different measuring teams originating from ten European Union member states.

The characteristics of installations, the conditions during field tests and the values of repeatability and reproducibility in the field are given in Annex D.

12 Equivalence with an alternative method

In order to show that an alternative method is equivalent to the reference method, follow the procedures described in the CEN/TS 14793.

The maximum allowable standard deviation of repeatability expressed in mg/m^3 for this reference method is:

$$s_{\text{rlimit}}(C) = 0,0119C + 0,01$$

where

C is the concentration in %.

The maximum allowable standard deviation of reproducibility expressed in mg/m^3 for this reference method is:

$$s_R(C) = 0,0103C + 0,0346$$

where

C is the concentration in %.

13 Test report

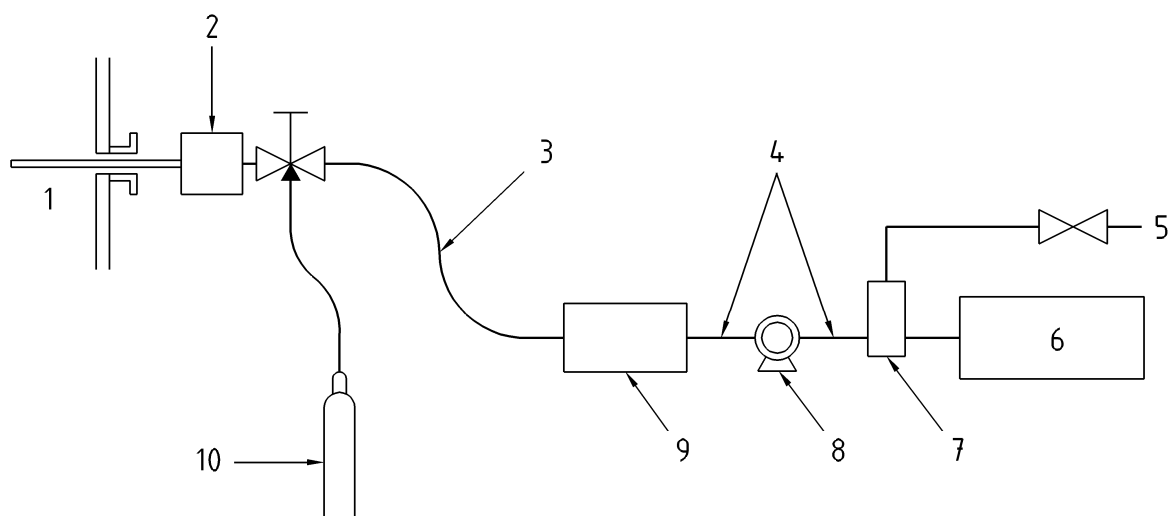
The test report shall include the following information:

- a) reference to this European standard;
- b) description of the purpose of tests;
- c) information about the personnel involved in the measurement;
- d) principle of gas sampling and conditioning;
- e) information about the analyser and description of the sampling and conditioning line;
- f) identification of the analyser used, and analyser's performance characteristics as listed in Table 1;
- g) operating range;
- h) details of the quality and the concentration of the calibration gases used;
- i) details on the adjustment performed before and after actual sampling (at the inlet of the sampling line and at the inlet of the analyser);
- j) description of plant and process;
- k) identification of the sampling plane;
- l) description of the location of the sampling point(s) in the sampling plane;
- m) description of the operating conditions of the plant process;
- n) changes in the plant operations during sampling, for example burner changes;
- o) measurement results with sampling date, time and duration;

- p) information on flue gas characteristics (temperature, velocity, moisture, pressure);
- q) time averaging on relevant periods;
- r) uncertainty budget established according to EN ISO 14956 or ENV 13005;
- s) any deviations from this European Standard.

Annex A (informative)

Schematic diagram of the measurement system



Key

- 1 Stack
- 2 Heated filter
- 3 Heated sample line
- 4 Sample gas transport line (PTFE)
- 5 Sample by-pass vent
- 6 Gas analyser
- 7 Sample gas manifold
- 8 Sample pump
- 9 Conditioning system: configuration 1: condenser with a cooling system – configuration 2: permeation drier
- 10 Calibration gas(es)

Figure A.1 - Schematic diagram of the measurement system

Annex B (informative)

Example of assessment of compliance of paramagnetic method for O₂ with requirements on emission measurements

B.1 Process of uncertainty estimation

B.1.1 General

This informative Annex gives an example of calculation of uncertainty budget established with configuration 1.

The procedure for calculating measurement uncertainty as follows is based on the law of propagation of uncertainty laid down in EN ISO 14956 or ENV 13005. The calculation procedure presents different steps:

B.1.2 Determination of model function

Define the measurand and all the parameters that influence the result of the measurement. These parameters, called "input quantities" shall be clearly defined.

Identify all sources of uncertainty contributing to any of the input quantities or to the measurand directly.

Then the model function, that is to say the relationship between the measurand and the influence quantities, shall be established, if possible in mathematical equation form.

B.1.3 Quantification of uncertainty components

Each uncertainty source is estimated to obtain its contribution to the overall uncertainty.

Use available performance characteristics of the measurement system, data from the dispersion of repeated measurements, data provided in calibration certificates.

Convert all uncertainty components (e.g. performance characteristics) to standard uncertainties of input and influence quantities.

B.1.4 Calculation of the combined uncertainty

Then the combined uncertainty u_c is calculated by combining standard uncertainties, by applying the "law of propagation of uncertainty".

In general, the uncertainty associated to a concentration is expressed in overall uncertainty form. The overall uncertainty U_c corresponds to the expanded combined uncertainty, obtained by multiplying by a coverage factor " k ": $U_c = k \times u_c$. The value of the coverage factor k is chosen on the basis of the level of confidence required. In most cases, k is taken equal to 2, for a level of confidence of approximately 95 %.

The following Tables give one example of:

- specific conditions of the site (Table B.1), that is to say the values and the variation range of the influence parameters;
- performance characteristics of the method (Table B.2) related to the parameters which can have an influence on the response of the analyser : the metrological performances of the analyser and the effect of influence parameters (environmental conditions like ambient temperature, voltage, pressure ... and chemical interferents).

B.2 Specific conditions in the site

Table B.1 — Example of site specific conditions

Specific conditions	Value/range
Range of analyser (O ₂)	0 % - 25 %
Mean of O ₂ concentration	12 % volume
Conditions in the field	
Sample volume flow variation	60 l/h $\pm$ 5 l/h
Temperature during adjustment	285 K
Fluctuations of ambient temperature during the measurement	283 K to 308 K
Voltage variation	230 V $\pm$ 5 %
Atmospheric pressure during the adjustment	99 kPa
Atmospheric pressure variation	99 kPa – 100 kPa
NO concentration variations in the field	100 mg/m ³ to 150 mg/m ³
NO ₂ concentration variations in the field	5 mg/m ³ to 7,5 mg/m ³
CO ₂ concentration variations in the field	8 % volume to 15 % volume
Calibration gas	20 % volume $\pm$ 1 % relative
O ₂ in N ₂ , without interferent	

B.3 Performance characteristics of the method

Table B.2 — Example of performance characteristics

Performance characteristics	Performance criteria	Results of laboratory and field tests
Response time	≤ 200 s	120 s
Detection limit	$\leq \pm 0,2$ %relative of the range	$\pm 0,1$ % of range
Lack of fit	$\leq \pm 0,3$ % vol	$\pm 0,2$ % vol
Zero drift	$\leq \pm 0,2$ %vol/24 h	$\pm 0,1$ %vol/24 h
Span drift	$\leq \pm 0,2$ %vol/24 h	$\pm 0,1$ %vol/24 h
Sensitivity to the sample volume flow	≤ 1 % of the range	0,8 % of the range/10 l/h
Sensitivity to the atmospheric pressure	$\leq 1,5$ % of the range/kPa	1,1 % of the range/kPa
Sensitivity to the ambient temperature	$\leq 0,3$ %volume/10 K	0,18 % volume/10 K
Sensitivity to the electric voltage at span	$\leq \pm 0,1$ %volume/10 V	0,08 % volume/10 V
Interferents	Total $\leq \pm 0,2$ %volume	
NO (300 mg/m ³)		0,05 % volume
NO ₂ (30 mg/m ³)		0,02 % volume
CO ₂ (10 %)		-0,003 % volume
Standard deviation of repeatability in laboratory at span level	≤ 0.4 % relative of the range	0,1 % of range

B.4 Calculation of the standard uncertainty of the concentration values given by the analyser

B.4.1 Model equation and application of the rule of the uncertainty propagation

The concentration in O₂ is equal to the concentration given by the analyser plus corrections due to deviations associated to influence quantities and to the performance characteristics of the analyser.

$$C_{O_2,\%} = C_{O_2,\text{read}} + \text{Corr}_{\text{fit}} + \text{Corr}_{0,\text{dr}} + \text{Corr}_{s,\text{dr}} + \text{Corr}_{\text{rep}} + \text{Corr}_{\text{adj}} + \text{Corr}_{\text{int}} + \sum_{j=1}^p \text{Corr}_{\text{inf}} \quad (\text{B.1})$$

where

$C_{O_2,\%}$: concentration of O₂ in % absolute;

$C_{O_2,\text{read}}$: concentration of O₂ given by the analyser;

Corr_{fit} : correction of lack of fit;

$\text{Corr}_{0,\text{dr}}, \text{Corr}_{s,\text{dr}}$: corrections of zero drift and span drift;

Corr_{rep} : correction of repeatability of the measurement;

Corr_{adj} : correction of adjustment;

Corr_{int} : correction of influence of interferents;

$\sum_{j=1}^p \text{Corr}_{\text{inf}}$: correction of influence quantities (ambient temperature (temp), atmospheric pressure (a,press), sample volume flow (s,vf), voltage (V)).

The corrections are specific for each analyser. Sometimes, value of correction is nil. For example, influence parameters (influence quantities: ambient temperature, atmospheric pressure... and interferents), can increase or decrease during the measurement period and are generally not monitored; therefore, we can consider that the best correction to be applied is zero.

Whether the corrections are equal to zero or not, the uncertainties associated to these corrections shall be taken into account in the calculation of the global uncertainty. The combined standard uncertainty associated to O₂ measurement is given by equation (2):

$$u^2(C_{O_2,\%}) = \sum_{i=1}^N \left[\left(\frac{\partial C_{O_2,\%}}{\partial X_i} \right)^2 u^2(X_i) \right] \quad (\text{B.2})$$

where

$u(C_{O_2,\%})$: is the combined standard uncertainty associated to O₂ measurement;

$X_i, i=1$ to N : are the influence quantities;

$\frac{\partial C_{O_2,\%}}{\partial X_i}$: is the sensitivity coefficient of $C_{O_2,\%}$ to X_i ;

$u(X_i)$ is the standard uncertainty associated to X_i ;

$\frac{\partial C_{O_2, \%}}{\partial X_i} u(X_i)$ is the partial uncertainty related to X_i .

The development of equation (2) gives:

$$\begin{aligned} u^2(C_{O_2, \%}) = & \left(\frac{\partial C_{O_2, \%}}{\partial C_{O_2, \text{read}}} \right)^2 u^2(C_{O_2, \text{read}}) + \left(\frac{\partial C_{O_2, \%}}{\partial \text{Corr}_{\text{fit}}} \right)^2 u^2(\text{Corr}_{\text{fit}}) + \left(\frac{\partial C_{O_2, \%}}{\partial \text{Corr}_{0, \text{dr}}} \right)^2 u^2(\text{Corr}_{0, \text{dr}}) \\ & + \left(\frac{\partial C_{O_2, \%}}{\partial \text{Corr}_{s, \text{dr}}} \right)^2 u^2(\text{Corr}_{s, \text{dr}}) + \left(\frac{\partial C_{O_2, \%}}{\partial \text{Corr}_{\text{rep}}} \right)^2 u^2(\text{Corr}_{\text{rep}}) + \left(\frac{\partial C_{O_2, \%}}{\partial \text{Corr}_{\text{int}}} \right)^2 u^2(\text{Corr}_{\text{int}}) + \left(\frac{\partial C_{O_2, \%}}{\partial \text{Corr}_{\text{temp}}} \right)^2 u^2(\text{Corr}_{\text{temp}}) \\ & + \left(\frac{\partial C_{O_2, \%}}{\partial \text{Corr}_{a, \text{press}}} \right)^2 u^2(\text{Corr}_{a, \text{press}}) + \left(\frac{\partial C_{O_2, \%}}{\partial \text{Corr}_{s, \text{vf}}} \right)^2 u^2(\text{Corr}_{s, \text{vf}}) + \left(\frac{\partial C_{O_2, \%}}{\partial \text{Corr}_V} \right)^2 u^2(\text{Corr}_V) + \left(\frac{\partial C_{O_2, \%}}{\partial \text{Corr}_{\text{adj}}} \right)^2 u^2(\text{Corr}_{\text{adj}}) \end{aligned} \quad (\text{B.3})$$

B.4.2 Calculation of the partial uncertainties

— Sensitivity coefficients:

$$\begin{aligned} \frac{\partial C_{O_2, \%}}{\partial C_{O_2, \text{read}}} = \frac{\partial C_{O_2, \%}}{\partial \text{Corr}_{\text{fit}}} = \frac{\partial C_{O_2, \%}}{\partial \text{Corr}_{0, \text{dr}}} = \frac{\partial C_{O_2, \%}}{\partial \text{Corr}_{s, \text{dr}}} = \frac{\partial C_{O_2, \%}}{\partial \text{Corr}_{\text{rep}}} = \frac{\partial C_{O_2, \%}}{\partial \text{Corr}_{\text{int}}} = \frac{\partial C_{O_2, \%}}{\partial \text{Corr}_{\text{temp}}} = \frac{\partial C_{O_2, \%}}{\partial \text{Corr}_{a, \text{press}}} \\ = \frac{\partial C_{O_2, \%}}{\partial \text{Corr}_{s, \text{vf}}} = \frac{\partial C_{O_2, \%}}{\partial \text{Corr}_V} = \frac{\partial C_{O_2, \%}}{\partial \text{Corr}_{\text{adj}}} = 1 \end{aligned} \quad (\text{B.4})$$

- $u(C_{O_2, \text{read}})$: the uncertainty related to the reading of the concentration is due to the resolution of the analyser and of the data acquisition. The uncertainty can be considered as negligible;
- $u(\text{Corr}_{\text{fit}})$: if $X_{\text{fit, max}}$ is the performance characteristic equal to the maximum deviation between the measured value and the value given through the linear regression achieved during the laboratory test, then, we can consider that the lack of fit has an equal probability to take any value included in the interval $[-X_{\text{fit, max}}; +X_{\text{fit, max}}]$. The standard uncertainty is calculated by application of a rectangular probability distribution:

$$u(\text{Corr}_{\text{fit}}) = \frac{X_{\text{fit, max}}/100 \times \text{range}}{\sqrt{3}} \quad (\text{B.5})$$

$$\frac{\partial C_{O_2, \%}}{\partial \text{Corr}_{\text{fit}}} u(\text{Corr}_{\text{fit}}) = \frac{X_{\text{fit, max}}/100 \times \text{range}}{\sqrt{3}} \quad (\text{B.6})$$

- $u(\text{Corr}_{0, \text{dr}})$, $u(\text{Corr}_{s, \text{dr}})$: where $X_{0, \text{dr}}$ and $X_{s, \text{dr}}$ are the zero and the span drift, we can suppose that the drifts have the same probability to take any value included in the intervals $[-X_{0, \text{dr}}; +X_{0, \text{dr}}]$ and $[-X_{s, \text{dr}}; +X_{s, \text{dr}}]$. The standard uncertainties are calculated by application of a rectangular probability distribution;

$$u(\text{Corr}_{0, \text{dr}}) = \frac{X_{0, \text{dr}}}{\sqrt{3}} \quad \text{and} \quad u(\text{Corr}_{s, \text{dr}}) = \frac{X_{s, \text{dr}}}{\sqrt{3}} \quad (\text{B.7})$$

$$\text{Then } \frac{\partial C_{O_2, \%}}{\partial \text{Corr}_{0, \text{dr}}} u(\text{Corr}_{0, \text{dr}}) = \frac{X_{0, \text{dr}}}{\sqrt{3}} \quad \text{and} \quad \frac{\partial C_{O_2, \%}}{\partial \text{Corr}_{s, \text{dr}}} u(\text{Corr}_{s, \text{dr}}) = \frac{X_{s, \text{dr}}}{\sqrt{3}} \quad (\text{B.8})$$

- $u(Corr_{rep})$: the standard uncertainty due to the repeatability is equal to the repeatability standard deviation calculated from the results of the repetitions of the measurements.

Several tests can be carried out at different concentrations. But only one of the values shall be included in the calculation:

- choose the repeatability standard deviation corresponding to the closest concentration measured in stack (in the example : the emission limit value);
- or the highest repeatability (relative) standard deviation whatever is the concentration measured in stack.

$$u(Corr_{rep}) = S_{rep}$$

where

S_{rep} : repeatability standard deviation

$$\text{Then } \frac{\partial C_{O_2, \%}}{\partial Corr_{rep}} u(Corr_{rep}) = S_{rep} \quad (B.9)$$

- Standard uncertainties associated to the influence quantities $u(Corr_{temp})$, $u(Corr_{a, press})$, $u(Corr_{s, vt})$, and $u(Corr_v)$: during laboratory test, the influence quantities are tested for one value of the parameter and the effects of the influence quantities are supposed to be proportional to the value of the parameter. The correction of the effect of an influence quantity is also proportional to its variation.

If X_j is an influence quantity: $Corr_{X_j} = c_j \times \Delta X_j$

where

c_j is a constant;

ΔX_j is the variation of the influence quantity.

The uncertainty associated to the correction is given by:

$$u^2(Corr_{X_j}) = \left(\frac{\partial Corr_{X_j}}{\partial \Delta X_j} \right)^2 u^2(\Delta X_j) \quad (B.10)$$

where

$\frac{\partial Corr_{X_j}}{\partial \Delta X_j} = c_j$ is the sensitivity coefficient; the sensitivity factor corresponds to the effect of the variation ΔX_j of the influence quantity in the response of the analyser; sensitivity coefficients are determined by laboratory tests;

$u(\Delta X_j)$ is the standard uncertainty associated to the variation of the influence quantity during the measuring period. We can suppose that the influence quantity has the same probability to take any value included in the interval ΔX_j .

In the standard EN ISO 14956, the calculation of the standard uncertainty associated to the variation of the influence quantity is given by the equation:

$$u(\Delta X_j) = \sqrt{\frac{(x_{j,\max} - x_{j,\text{adj}})^2 + (x_{j,\min} - x_{j,\text{adj}}) \times (x_{j,\max} - x_{j,\text{adj}}) + (x_{j,\min} - x_{j,\text{adj}})^2}{3}} \quad (\text{B.11})$$

where

$X_{j,\min}$ is the minimum value of the influence quantity X_j during the measuring period;

$X_{j,\max}$ is the maximum value of the influence quantity X_j during the measuring period;

$X_{j,\text{adj}}$ is the value of the influence quantity X_j during the adjustment of the analyser.

$$\text{Then } \frac{\partial C_{O_2, \%}}{\partial \text{Corr}_{X_j}} u(\text{Corr}_{X_j}) = \frac{\partial \text{Corr}_{X_j}}{\partial X_j} u(\Delta X_j)$$

$$\frac{\partial C_{O_2, \%}}{\partial \text{Corr}_{X_j}} u(\text{Corr}_{X_j}) = c_j \sqrt{\frac{(x_{j,\max} - x_{j,\text{adj}})^2 + (x_{j,\min} - x_{j,\text{adj}}) \times (x_{j,\max} - x_{j,\text{adj}}) + (x_{j,\min} - x_{j,\text{adj}})^2}{3}} \quad (\text{B.12})$$

The calculation can be simplified in the two following cases:

— First case : if the value $X_{j,\text{adj}}$ is at the centre of the interval $[X_{j,\max}; X_{j,\min}]$:

$$u(\Delta X_j) = \frac{(x_{j,\max} - x_{j,\min})}{\sqrt{12}} \quad \text{and} \quad \frac{\partial C_{O_2, \%}}{\partial \text{Corr}_{X_j}} u(\text{Corr}_{X_j}) = c_j \frac{(x_{j,\max} - x_{j,\min})}{\sqrt{12}} \quad (\text{B.13})$$

— Second case : if the value $X_{j,\text{adj}}$ is equal to $X_{j,\max}$ or $X_{j,\min}$:

$$u(\Delta X_j) = \frac{(x_{j,\max} - x_{j,\min})}{\sqrt{3}} \quad \text{and} \quad \frac{\partial C_{O_2, \%}}{\partial \text{Corr}_{X_j}} u(\text{Corr}_{X_j}) = c_j \frac{(x_{j,\max} - x_{j,\min})}{\sqrt{3}} \quad (\text{B.14})$$

— Standard uncertainties associated to the interferences: the case of the interferences is similar to the case of influence quantities. The effect of the interferences is tested for one concentration of interferent and is supposed to be proportional to the value of the parameter. The correction of the effect of an influence quantity is also proportional to its variation.

If Int_j is an interferent for the measured compound, and ΔInt_j its variation interval during the measurement period, the standard uncertainty of the correction is given by:

$$u^2(\text{Corr}_{\text{Int}_j}) = \left(\frac{\partial \text{Corr}_{\text{Int}_j}}{\partial \Delta \text{Int}_j} \right)^2 u^2(\Delta \text{Int}_j) \quad (\text{B.15})$$

where

$\frac{\partial \text{Corr}_{\text{Intj}}}{\partial \Delta \text{Int}_j}$ is the sensitivity coefficient ; sensitivity coefficients are determined by laboratory tests.

$u(\Delta \text{Int}_j)$ is the standard uncertainty associated to the variation of the concentration of the interferent Int_j during the measuring period. We can suppose that the influence quantity has the same probability to take any value included in the interval ΔInt_j .

In the standard EN ISO 14956, the calculation of the standard uncertainty associated to the variation of the influence quantity is given by the equation:

$$u(\Delta \text{Int}_j) = \sqrt{\frac{(\text{Int}_{j,\text{max}} - \text{Int}_{j,\text{adj}})^2 + (\text{Int}_{j,\text{min}} - \text{Int}_{j,\text{adj}}) \times (\text{Int}_{j,\text{max}} - \text{Int}_{j,\text{adj}}) + (\text{Int}_{j,\text{min}} - \text{Int}_{j,\text{adj}})^2}{3}} \quad (\text{B.16})$$

where

$\text{Int}_{j,\text{min}}$ is the minimum concentration of the interferent j during the measuring period;

$\text{Int}_{j,\text{max}}$ is the maximum concentration of the interferent j during the measuring period;

$\text{Int}_{j,\text{adj}}$ is the concentration of the interferent j in the calibration gas used to adjust the analyser;

$$\text{Then } \frac{\partial \text{CO}_2, \%}{\partial \text{Corr}_{\text{Intj}}} u(\text{Corr}_{\text{Intj}}) = \left(\frac{\partial \text{Corr}_{\text{Intj}}}{\partial \Delta \text{Int}_j} \right) u(\Delta \text{Int}_j) \quad (\text{B.17})$$

$$\frac{\partial \text{CO}_2, \%}{\partial \text{Corr}_{\text{Intj}}} u(\text{Corr}_{\text{Intj}}) = \frac{c_j}{\text{Int}_{j,\text{test}}} \times \sqrt{\frac{(\text{Int}_{j,\text{max}} - \text{Int}_{j,\text{adj}})^2 + (\text{Int}_{j,\text{min}} - \text{Int}_{j,\text{adj}}) \times (\text{Int}_{j,\text{max}} - \text{Int}_{j,\text{adj}}) + (\text{Int}_{j,\text{min}} - \text{Int}_{j,\text{adj}})^2}{3}}$$

where

$c_j = \frac{\partial \text{Corr}_{\text{Intj}}}{\partial \Delta \text{Int}_j}$ is the sensitivity coefficient;

$\text{Int}_{j,\text{test}}$ is the concentration of the interferent used to determine the sensitivity coefficient c_j during laboratory test.

The calculation can be simplified if the value $X_{j,\text{adj}}$ is equal to zero, that is to say that the concentration of the interferent in the calibration gas is equal to zero, what is often the case. The standard uncertainty is given by:

$$u(\Delta \text{Int}_j) = \sqrt{\frac{\text{Int}_{j,\text{max}}^2 + \text{Int}_{j,\text{min}} \times \text{Int}_{j,\text{max}} + \text{Int}_{j,\text{min}}^2}{3}} \quad \text{and} \quad (\text{B.18})$$

$$\frac{\partial \text{CO}_2, \%}{\partial \text{Corr}_{\text{Intj}}} u(\text{Corr}_{\text{Intj}}) = \frac{c_j}{\text{Int}_{j,\text{test}}} \times \sqrt{\frac{\text{Int}_{j,\text{max}}^2 + \text{Int}_{j,\text{min}} \times \text{Int}_{j,\text{max}} + \text{Int}_{j,\text{min}}^2}{3}}$$

All interferences shall not be included in the calculation of the global uncertainty. In accordance with the standard EN ISO 14956, the calculation shall include the highest value between the sum $S_{\text{int,p}}$ of the interferences with positive impact in the response of the analyser and the sum $S_{\text{int,n}}$ of the interferences with negative impact.

$$S_{\text{int,p}} = \frac{\partial C_{\text{O}_2, \%}}{\partial \text{Corr}_{\text{int,p1}}} u(\text{Corr}_{\text{int,p1}}) + \frac{\partial C_{\text{O}_2, \%}}{\partial \text{Corr}_{\text{int,p2}}} u(\text{Corr}_{\text{int,p2}}) + \dots + \frac{\partial C_{\text{O}_2, \%}}{\partial \text{Corr}_{\text{int,pk}}} u(\text{Corr}_{\text{int,pk}}) \quad (\text{B.19})$$

$$S_{\text{int,n}} = \frac{\partial C_{\text{O}_2, \%}}{\partial \text{Corr}_{\text{int,n1}}} u(\text{Corr}_{\text{int,n1}}) + \frac{\partial C_{\text{O}_2, \%}}{\partial \text{Corr}_{\text{int,n2}}} u(\text{Corr}_{\text{int,n2}}) + \dots + \frac{\partial C_{\text{O}_2, \%}}{\partial \text{Corr}_{\text{int,nt}}} u(\text{Corr}_{\text{int,nt}})$$

where

$S_{\text{int,p}}$ is the sum of the interferences with positive impact;

$\frac{\partial C_{\text{O}_2, \%}}{\partial \text{Corr}_{\text{int,p1}}} u(\text{Corr}_{\text{int,p1}})$ is the partial uncertainty due to the 1st interferent with positive impact;

$\frac{\partial C_{\text{O}_2, \%}}{\partial \text{Corr}_{\text{int,pk}}} u(\text{Corr}_{\text{int,pk}})$ is the partial uncertainty due to the k^{th} interferent with positive impact;

$S_{\text{int,n}}$ is the sum of the interferences with negative impact;

$\frac{\partial C_{\text{O}_2, \%}}{\partial \text{Corr}_{\text{int,n1}}} u(\text{Corr}_{\text{int,n1}})$ is the partial uncertainty due to the k^{th} interferent with negative impact;

$\frac{\partial C_{\text{O}_2, \%}}{\partial \text{Corr}_{\text{int,nt}}} u(\text{Corr}_{\text{int,nt}})$ is the partial uncertainty due to the t^{th} interferent with negative impact.

$$\left(\frac{\partial C_{\text{O}_2, \%}}{\partial \text{Corr}_{\text{int}}} \right)^2 u^2(\text{Corr}_{\text{int}}) = \max[S_{\text{int,p}}^2; S_{\text{int,n}}^2]$$

- $u(\text{Corr}_{\text{adj}})$: standard uncertainty $u(\text{Corr}_{\text{adj}})$ is calculated from the uncertainty of the calibration gas. In general the uncertainty given by manufacturer is the expanded uncertainty. If X_{cal} is the expanded uncertainty of the calibration gas expressed in %, the standard uncertainty of the correction at the oxygen concentration C_{O_2} is:

$$u(\text{Corr}_{\text{adj}}) = \frac{X_{\text{cal}}/100 \times C_{\text{O}_2}}{2} \quad (\text{B.20})$$

$$\text{Then } \frac{\partial C_{\text{O}_2, \%}}{\partial \text{Corr}_{\text{adj}}} u(\text{Corr}_{\text{adj}}) = \frac{X_{\text{cal}}/100 \times C_{\text{O}_2}}{2}$$

B.4.3 Results of uncertainty calculation

B.4.3.1 Standard uncertainties

Table B.3 — Results of uncertainty calculation

Performance characteristic	Partial standard uncertainty	Value of partial standard uncertainty at limit value (in % volume)
Lack of fit	$U(Corr_{fit})$	$\frac{0,2}{\sqrt{3}}=0,116$
Zero drift	$U(Corr_{0,dr})$	$\frac{0,1}{\sqrt{3}}=0,058$
Span drift	$u(Corr_{s,dr})$	$\frac{0,1}{\sqrt{3}}=0,058$
Sensitivity to the sample volume flow	$u(Corr_{s,vf})$	$\frac{(0,8/100)}{10} \times 25 \frac{10}{\sqrt{3}}=0,116$
Sensitivity to atmospheric pressure	$u(Corr_{a,press})$	$(1,1/100) \times 25 \frac{100-99}{\sqrt{3}}=0,159$
Sensitivity of ambient temperature	$u(Corr_{temp})$	$\frac{0,18}{10} \times \sqrt{\frac{(308-285)^2 + (308-285)(283-285) + (283-285)^2}{3}}=0,230$
Sensitivity of electric voltage	$u(Corr_{volt})$	$\frac{0,08}{10} \times \frac{(230+230/100 \times 5) - (230-230/100 \times 5)}{\sqrt{12}}=0,054$
Interferent : NO	$u(Corr_{NO})$	$\frac{0,05}{300} \times \sqrt{\frac{150^2 + 150 \times 100 + 100^2}{3}}=0,021$
Interferent : NO ₂	$u(Corr_{NO2})$	$\frac{0,02}{30} \times \sqrt{\frac{7,5^2 + 7,5 \times 5 + 5^2}{3}}=0,005$
Interferent : CO ₂	$u(Corr_{CO2})$	$\frac{-0,003}{10} \times \sqrt{\frac{8^2 + 8 \times 15 + 15^2}{3}}=-0,004$
Repeatability standard deviation at span level	$u(Corr_{rep})$	$(0,1/100) \times 25=0,025$
Uncertainty of calibration gas	$u(Corr_{adj})$	$\frac{(1/100) \times 12}{2}=0,06$

B.4.3.1.1 Combined uncertainty

$$S_{int,p} > S_{int,n}$$

$$S_{int,p} = \frac{\partial C_{O_2, \%}}{\partial Corr_{NO}} u(Corr_{NO}) + \frac{\partial C_{O_2, \%}}{\partial Corr_{NO2}} u(Corr_{NO2}) = 0,026 \text{ \% vol} \quad (B.21)$$

$$u(C_{O_2,\%}) = \sqrt{u^2(Corr_{fit}) + u^2(Corr_{O,dr}) + u^2(Corr_{S,dr}) + u^2(Corr_{rep}) + u^2(Corr_{adj}) + u^2(Corr_{S,vf}) + u^2(Corr_{a,press}) + u^2(Corr_{temp}) + u^2(Corr_{volt}) + S_{int,p}^2}$$

$$u(C_{O_2,\%}) = \sqrt{0,116^2 + 0,058^2 + 0,058^2 + 0,025^2 + 0,06^2 + 0,116^2 + 0,159^2 + 0,230^2 + 0,054^2 + (0,021 + 0,005)^2}$$

$$u(C_{O_2,\%}) = 0,35 \text{ \% volume}$$

B.4.3.1.2 Overall uncertainty

$$U(C_{O_2,\%}) = \pm 0,69 \text{ \% volume (k=2)}$$

$$U(C_{O_2,\%}) = \pm 5,8 \text{ \% relative (k=2)}$$

B.4.3.1.3 Evaluation of the compliance with the required measurement quality

Interferents with a positive impact:

$$\text{Total} = 0,07 \text{ \% vol} \quad \text{Total} < 0,2 \text{ \% volume}$$

Interferents with a negative impact:

$$\text{Total} = 0,003 \text{ \% vol} \quad \text{Total} < 0,2 \text{ \% volume}$$

All values of the performance characteristics obtained in the laboratory and field tests are complying with the requirements. Conclusion: the measurement method fulfils the requirements.

Annex C (informative)

Procedure of correction of data from drift effect

The drift is supposed to be proportional to the time.

B C D E F G H

Adjustment and span check after measurement				
	Gas	expected value	reading before	reading after
	Span	918,9	912,3	882,4
	zero	0	1,8	1,7
	Time		5,36	17,58
	duration		F7-E7	
	In minutes		742	

Fill in only boxes with shaded area

Relation between the true value (Y) and the reading (X), Y=AX+B				
Coefficients	Adjustment	Check	deviation	Drift / min.
A (span)	(D5-D6)/(E5-E6)	(D5-D6)/(F5-F6)	E13-D13	(E13-D13)/E9
B(zero)	D13*-E6	E13*-F6	E14-D14	(E14-D14)/E9

Drift is $1 - ((D5 - (D5 * F13 + F14)) / D5)$

Equation to calculate the corrected concentration (C cor.) from the reading (C) and time (t)

$$C \text{ cor.} = C \times (C14 + E14 \times t) + (C15 + E15 \times t)$$

For this application :

CONCATENER("C cor. = C x (","CTXT(D13;4);" + ","CTXT(G13;4);" x t) + (","CTXT(D14;4);" + ","CTXT(G14;4);" x t)"))

Example		
C	Time	C cor.
E5	0	$D28 * (D\$13 + G\$13 * E28) + (D\$14 + G\$14 * E28)$
F5	E9	$D29 * (D\$13 + G\$13 * E29) + (D\$14 + G\$14 * E29)$
180	124	$D30 * (D\$13 + G\$13 * E30) + (D\$14 + G\$14 * E30)$

Annex D (informative)

Evaluation of the method in the field

D.1 General

The method has been evaluated during six field tests, on waste incineration installations, co-incineration installations and large combustion plants. Each test was performed by at least four different European measuring teams originating from ten member countries.

D.2 Characteristics of installations

- 1st field test : INERIS bench-loop at Verneuil en Halatte (F) ; the bench-loop simulates combustion or waste incineration exhaust gases. Five teams took part in the 1st field test. Double measurements were not performed simultaneously but sequentially. Five different flue gas matrices were generated. Within each matrix, two sequential measurements were performed. Two additional sequential measurements were performed in flue gas matrices where the flue gas concentrations varied. There were a total of twelve measurements performed by all the teams.
- 2nd field test: waste incinerator in Denmark. Four teams took part to the field test and performed double measurements simultaneously. A total of sixteen measurements was performed by all the teams:
- 3rd field test: waste incinerator in Italy. Four teams took part to the field test. Two pairs of two teams performed double measurements simultaneously and the four teams performed single measurements simultaneously. A total of six double measurements was performed by each pair of two teams while a total of twelve single measurements was performed by all teams.
- 4th field test: co-incinerator combined heat and power installation in Sweden. Two pairs of two teams performed double measurements simultaneously and the four teams performed single measurements simultaneously. A total of six double measurements was performed by each pair of two teams while a total of twelve single measurements was performed by all the teams.
- 5th field test: co-incinerator cement plant in Germany. The fuel could be coal, heavy oil and secondary fuel (e.g. paper, plastics, textiles, and tires). Four teams took part to the field test and performed double measurements simultaneously. A total of sixteen double measurements was performed by all the teams.
- 6th field test: coal fired power plant in Germany. Four teams performed their double measurements simultaneously. The total amount of double measurements performed by all teams was 12.

An overview of the flue gas characteristics is given in Table D.1.

Table D.1 — Flue gas characteristics during field tests

Field test	Installation	Fuel	Flue gas characteristics						
			T °C	O ₂ % vol	NO _x mg/(n)m ³	SO ₂ mg/(n)m ³	CO mg/(n)m ³	H ₂ O % vol	PM mg/(n)m ³
1	Power plant ^a	Natural gas	< 150	5 to 13	10 to 1 300	10 to 2 000	20 to 400	10 to 21	< 1
2	Waste incinerator	Municipal waste	90 to 110	8 to 11	180 to 250	25 to 250	5 to 15	13 to 19	1 to 5
3	Waste incinerator	Municipal waste	85 to 105	16 to 18	61 to 78	5 to 50	0 to 2	8 to 12	1 to 5
4	Co-incinerator	Wood, waste, coal	70 to 80	4 to 6	4 to 70	0 to 10	50 to 150	8 to 12	0 to 20
5	Co-incinerator	Coal, oil, waste	140 to 170	4 to 6	440 to 1060	60 to 170	260 to 740	23 to 26	5 to 10
6	Power plant	Coal	130 to 140	8,9 to 9,2	110 to 140	1 000 to 1 130	3 to 6	5,5 to 8	< 50

^a Bench-loop: flue gas simulation.

D.3 Repeatability and reproducibility in the field

Repeatability standard deviation s_r and reproducibility standard deviation s_R are determined by performing inter-laboratory tests.

Repeatability standard deviation s_r , internal confidence interval (CI_r) and repeatability in the field r are calculated according to ISO 5725-2 and ISO 5725-6, from the results of the double measurements implemented by the same laboratory.

$$CI_r = t_{0,95;n-1} \times s_r \text{ and } r = \sqrt{2} \times t_{0,95;n-1} \times s_r \quad (D.1)$$

where

CI_r is the internal confidence interval;

s_r is the repeatability standard deviation;

$t_{0,95;n-1}$ is the student factor for a level of confidence of 95 % and a degree of freedom of $n-1$ (n : number of double measurements);

r is the repeatability in the field.

Reproducibility standard deviation s_R , external confidence interval (CI_R) and reproducibility in the field R are calculated according to ISO 5725-2, from the results of parallel measurements performed simultaneously by several laboratories.

$$CI_R = t_{0,95;np-1} \times s_R \text{ and } R = \sqrt{2} \times t_{0,95;np-1} \times s_R \quad (\text{D.2}) \text{ where}$$

CI_R is the external confidence interval;

s_R is the repeatability standard deviation;

$t_{0,95;np-1}$ is the student factor for a level of confidence of 95 % and a degree of freedom of np-1 (n : number of measurements; p: number of laboratories);

R is the reproducibility in the field.

Table D.2 — Repeatability in the field

Field test	Concentration		Number of teams	Number of double measurements	Repeatability standard deviation	95 % confidence interval for a single measurement CI_r	Repeatability	
					s_r		r	
	Range% volume	Average% volume			% volume	% volume	% volume	% relative
1A		11,2	5	2	0,018	0,041	0,058	0,5
1B		12,6	5	2	0,018	0,041	0,058	0,5
1C		9,2	5	2	0,017	0,038	0,054	0,6
1D		6,1	5	2	0,015	0,034	0,048	0,8
1E		3,1	5	2	0,013	0,030	0,042	1,4
1F	6,8 to 7,2	7,1	5	2	0,016	0,035	0,050	0,7
2	8,5 to 10,5	9,2	4	16	0,09	0,20	0,28	3,1
3	16,4 to 17,7	17	4	12	0,19	0,41	0,58	3,4
4	4,5 to 5,3	4,8	4	12	0,04	0,09	0,13	2,7
5	4,3 to 6,0	4,9	3	12	0,05	0,10	0,14	2,9
6	8,9 to 9,2	9,1	4	12	0,08	0,17	0,24	2,7

The following functions were determined:

$$s_r(C) = 0,0119 C + 0,01 \quad \text{in \%vol;}$$

$$s_{r \text{ limit}}(C) = 0,0119 C + 0,01 \quad \text{in \%vol;}$$

$$CI_r(C) = 0,022 C + 0,03 \quad \text{in \%vol;}$$

$$r(C) = 0,031 C + 0,042 \quad \text{in \%vol.}$$

Table D.3 — Reproducibility in the field

Field test	Concentration		Number of teams	Number of measurements	Reproducibility standard deviation s_R % volume	95 % confidence interval for a single measurement CI_R % volume	Reproducibility R	
	Range % volume	Average % volume					% volume	% relative
1A		11,2	5	2	0,058	0,13	0,18	1,6
1B		12,6	5	2	0,06	0,13	0,19	1,5
1C		9,2	5	2	0,06	0,13	0,19	2,1
1D		6,1	5	2	0,041	0,093	0,13	2,1
1E		3,1	5	2	0,027	0,061	0,09	2,8
1F	6,8 to 7,2	7,1	5	2	0,12	0,27	0,39	5,4
2	8,5 to 10,5	9,3	4	16	0,11	0,23	0,33	3,5
3	16,4 to 17,7	17	4	12	0,22	0,49	0,69	4,1
4	4,7 to 5,2	5	4	12	0,13	0,29	0,41	8,2
5	4,3 to 6,0	4,9	3	12	0,06	0,13	0,18	3,8
6	8,9 to 9,2	9,1	4	12	0,12	0,26	0,37	4,1

The following functions were determined:

$$s_R(C) = 0,0103 C + 0,0346 \quad \text{in \%vol;}$$

$$CI_R(C) = 0,029 C + 0,07 \quad \text{in \%vol;}$$

$$R(C) = 0,041 C + 0,1 \quad \text{in \%vol.}$$

Annex ZA (informative)

Relationship between this European Standard and the Essential Requirements of EU Directive

This European Standard has been prepared under a mandate given to CEN by the European Commission and the European Free Trade Association and supports essential requirements of the EU Council Directive 2000/76/EC on the Incineration of Hazardous Waste and of the EU Council Directive 2001/80/EC on the limitation of emissions of certain pollutants into the air from large combustion plants.

WARNING — Other requirements and EU Directives may be applicable falling within the scope of use of this European Standard.

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